

10-701 Machine Learning
Fall 2024
PRACTICE Exam 1
9/30/2024
Time Limit: xxx minutes

Name:

AndrewID:

Instructions:

- Fill in your name and Andrew ID above. Be sure to write neatly, or you may not receive credit for your exam.
- This exam contains 21 pages (including this cover page).
There are 8 sections and the total number of points is 68.
- Clearly mark your answers in the allocated space in pen or **darkly in pencil**: your exams will be scanned for electronic grading and writing too faintly may make your answers illegible.
- Look over the exam first to make sure that none of the 21 pages are missing. The problems are of varying difficulty, so you may wish to pick off the easy ones first.
- The last few pages are left blank for you to use for scratch work. Please do **not** remove these pages!
- No electronic devices may be used during the exam.
- You have xxx minutes to complete the exam. Good luck!

Topic	Section	Points
Decision Trees	1	x
Linear Separators	2	x
MLE/MAP	3	x
Logistic Regression	4	x
Regularization	5	x
Kernels	6	x
Learning Theory	7	x
—	Total	x

Instructions for Specific Problem Types

For “Select One” questions, please fill in the appropriate bubble completely:

Select One: Who taught this course?

- ☒ Henry Chai
- ☐ Marie Curie
- ☐ Noam Chomsky

If you need to change your answer, you may cross out the previous answer and bubble in the new answer:

Select One: Who taught this course?

- ☒ Henry Chai
- ☐ Marie Curie
- ☒ Noam Chomsky

For “Select all that apply” questions, please fill in all appropriate squares completely:

Select all that apply: Which are scientists?

- ☒ Stephen Hawking
- ☒ Albert Einstein
- ☒ Isaac Newton
- ☐ I don't know

Again, if you need to change your answer, you may cross out the previous answer(s) and bubble in the new answer(s):

Select all that apply: Which are scientists?

- ☒ Stephen Hawking
- ☒ Albert Einstein
- ☒ Isaac Newton
- ☒ I don't know

For questions where you must fill in a blank, please make sure your final answer is fully included in the given space. You may cross out answers or parts of answers, but the final answer must still be within the given space.

Fill in the blank: What is the course number?

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1 Decision Trees

1. Suppose you have a training dataset consisting of 3 boolean features X_1, X_2 , and X_3 where $X_i \in \{0, 1\}$. Furthermore, assume the label is defined as $Y = X_1 \vee X_2$ i.e., $Y = 1$ if $X_1 = 1$ or $X_2 = 1$ and $Y = 0$ otherwise. Suppose that your training dataset contains all of the 8 possible feature vectors:

X_1	X_2	X_3	Y
0	0	0	0
1	0	0	1
0	1	0	1
1	1	0	1
0	0	1	0
1	0	1	1
0	1	1	1
1	1	1	1

- (a) What is the training error rate of a depth-0 decision tree on this dataset?

- (b) If your splitting criterion is training error rate, which feature or features would you split on first? Briefly justify your answer.

- (c) If your splitting criterion is information gain, which feature or features would you split on first? Briefly justify your answer.

2. You are given a dataset for binary classification with two features x_1 and x_2 . x_1 can have two possible values 0,1 and x_2 can have three possible values 0,1,2. Figure 1 provides a depiction of this dataset.

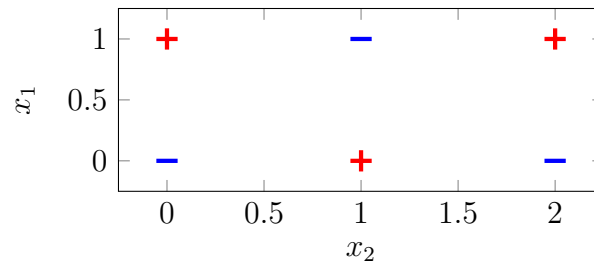


Figure 1: Binary classification dataset.

- (a) What is the lowest possible training error rate any decision tree could achieve on this dataset? The tree need not be a binary tree but each node should split on at most one feature.

- (b) Draw a decision tree that achieves this training error rate.

3. We would like to learn a decision tree. We have n samples for training. You can assume that n is large and we are using continuous features. In the following questions, suppose we only use the first feature, x_1 .

(a) We would like to split according to x_1 at the root with 3 branches: data points are split at the root to three different subtrees by findings values a and b such that the three subtrees are $x_1 \leq a$, $a < x_1 < b$ and $x_1 \geq b$. What is the runtime for finding the values of x_1 that should be used by the root for such a split?

☐ $O(\log n)$

☐ $O(\sqrt{n})$

☐ $O(n)$

☐ $O(n^2)$

☐ $O(n^3)$

(b) Following our first split, we would like to *split again* on x_1 in each of the three sub-trees resulting from the split in the previous question. Again, for each subtree we would split three ways as we did in the root. What is the total runtime for determining the optimal splits for ALL three subtrees?

☐ $O(n)$

☐ $O(n^2)$

☐ $O(n^3)$

☐ $O(n^4)$

☐ $O(n^8)$

2 Linear Separators

1. (1 point) Recall that a linear separator in 2 dimensions is given a function of the form:

$$y = \phi(w_1x_1 + w_2x_2 + b)$$

where $y, x_i \in \{0, 1\}$, and transfer function $\phi: \mathcal{R} \rightarrow \{0, 1\}$

$$\phi(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ 1 & \text{otherwise} \end{cases}$$

- (a) (1 point) Complete the following truth table for the NAND (NOT AND, i.e. NOT operation applied to the logical AND of the two inputs) operation.

x_1	x_2	NAND
0	0	
0	1	
1	0	
1	1	

- (b) (3 points) Provide values of w_1, w_2 and b to use the linear separator for logical NAND.

$w_1 =$

$w_2 =$

$b =$

- (c) (1 point) Complete the following truth table for the NOR (NOT OR, i.e. NOT operation applied to the logical OR of the two inputs) operation.

x_1	x_2	NOR
0	0	
0	1	
1	0	
1	1	

- (d) (3 points) Provide values of w_1, w_2 and b to use the linear separator for logical NOR.

$w_1 =$

$w_2 =$

$b =$

- (e) (1 point) Complete the following truth table for the XOR operation.

x_1	x_2	XOR
0	0	
0	1	
1	0	
1	1	

- (f) (6 points) Prove that the linear separator ϕ cannot be used to create logical XOR.

3 MLE/MAP

1. **True or False:** Recall that the MLE of a parameter θ given some dataset $\mathcal{D}$ is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D}|\theta)$$

while the MAP estimate of θ is

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\mathcal{D}|\theta)p(\theta).$$

For every dataset, assuming that the parameter space is bounded, there will always exist some prior distribution $p(\theta)$ for which the MAP estimate is equivalent to the MLE.

☐ True

☐ False

2. The probability density function of the Pareto distribution is

$$p(x|k, \alpha) = \begin{cases} \frac{\alpha k^\alpha}{x^{\alpha+1}} & x \in [k, \infty) \\ 0 & \text{otherwise} \end{cases}$$

$k, \alpha \in (0, \infty)$

Given n independent samples $x_1, x_2, \dots, x_n$ drawn from a Pareto distribution, what is the MLE for the parameters k and α ? **Hint: first determine the MLE of k , then use that result to in your expression for the MLE of α .**

$$\hat{k}_{MLE} = \underline{\hspace{2cm}}$$

$$\hat{\alpha}_{MLE} = \underline{\hspace{2cm}}$$

4 Logistic Regression

1. Consider a dataset with only binary features, $\mathbf{x} \in \{0, 1\}^d$, where feature x_1 is rarely 1 and only takes on value 1 in the training dataset if the corresponding label is 1. If you fit a logistic regression model to this dataset using gradient descent, what value will the coefficient on this feature, $\hat{w}_1$, approach? Briefly justify your answer.

2. A generalization of logistic regression to a multiclass settings involves expressing the per-class probabilities $P(y = c|x)$ as the softmax function $\frac{\exp(w_c^T x)}{\sum_{d \in C} \exp(w_d^T x)}$, where c is some class from the set of all classes C and w_c is a class-specific weight vector.

Consider a binary classification problem with labels encoded as $y \in \{0, 1\}$. Rewrite the expression above for this situation to end up with expressions for $P(Y = 1|x)$ and $P(Y = 0|x)$ that we derived in class for binary logistic regression.

5 Regularization

1. Complete the following paragraph about L_1 and L_2 regularization by circling the best of the provided options for each of blanks:

When applied to linear regression, L_1 regularization tends to learn

denser / sparser weight vectors than L_2 regularization because the

constraint surface is non-convex / square-shaped. The optimal

weight vector for linear regression with L_1 regularization can / cannot

be solved for in closed form.

6 Kernels

Consider the following kernel function:

$$K(x, x') = \begin{cases} 1, & \text{if } x = x' \\ 0, & \text{otherwise} \end{cases}$$

1. **[9 pts]** Prove this is a valid kernel. That is, describe an implicit mapping $\Phi : X \rightarrow \mathcal{R}^m$ such that $K(x, x') = \Phi(x) \cdot \Phi(x')$. (You may assume the input space X is finite.)

2. **[9 pts]** In this kernel space, any labeling of points in X will be linearly separable. Justify this claim.

3. [9 pts] Since all labelings are linearly separable, this kernel seems perfect for learning any target function. Why is this actually a bad idea?

4. [9 pts] Assume $k_1(\cdot, \cdot)$ is a kernel with corresponding feature mapping $\phi_1 : \mathbb{R}^M \rightarrow \mathbb{R}^{M_1}$, and $k_2(\cdot, \cdot)$ is a kernel with corresponding feature mapping $\phi_2 : \mathbb{R}^M \rightarrow \mathbb{R}^{M_2}$, both acting on the same space. Prove that, $k'(\mathbf{x}, \mathbf{z}) = 2(k_1(\mathbf{x}, \mathbf{z}))^2 + 3(k_2(\mathbf{x}, \mathbf{z}))^2 + 4k_1(\mathbf{x}, \mathbf{z})k_2(\mathbf{x}, \mathbf{z}) + k_1(\mathbf{x}, \mathbf{z}) + k_2(\mathbf{x}, \mathbf{z})$ is also a valid kernel.

5. **Short answer:** In 2-3 concise sentences, briefly describe the primary benefit of the kernel trick.

6. **Select all that apply:** If k is a valid kernel, which of the following statements must be true?

- ☐ $k(x, x') = k(x', x) \forall x$ and x'
- ☐ $k(x, x') \geq 0 \forall x$ and x'
- ☐ If $k(x, x) = k(x', x')$, then $x = x'$
- ☐ $k(x, x') = k(x - x', 0) \forall x$ and x'
- ☐ None of the above

7 Learning Theory

1. **Fill in the Blanks:** Complete the following sentence by circling one option in each square:

In order to prove that the VC-dimension of a hypothesis set $\mathcal{H}$ is D , you must

show that $\mathcal{H}$ can / cannot shatter any set / some set / multiple sets

of D data points and can / cannot shatter any set / some set / multiple sets

of $D + 1$ data points.

2. **Math:** For an arbitrary finite hypothesis set $\mathcal{H}$, provide an upper bound on the VC-dimension of $\mathcal{H}$ in terms of $|\mathcal{H}|$.

3. Let $\mathcal{X} = \mathbb{R}^2$ and $\mathcal{Y} = \{+1, -1\}$. Let

$$H = \{h_{a_1, a_2} \mid a_1, a_2 \in \mathbb{R}, h_{a_1, a_2}(x_1, x_2) = +1 \text{ iff } x_1 \leq a_1 \text{ and } x_2 \leq a_2\},$$

be the hypothesis class corresponding to positive “quarter planes” in $\mathbb{R}^2$

- (a) **Math:** Show that the VC dimension of H is at least 2.

- (b) **Math:** Show that the VC dimension of H is at most 2.

- (c) **Math:** Verify the Sauer-Shelah lemma for H by showing that the number of possible distinct labellings produced by H on m points is $O(m^2)$.

4. Let X be the feature space and let D be a distribution over X . We have training samples $S : (x_1, c^*(x_1)), \dots, (x_m, c^*(x_m))$, x_i i.i.d from D . We assume labels $c^*(x_i) \in \{-1, 1\}$.

Let $\mathcal{H}$ be a hypothesis space and let $h \in \mathcal{H}$ be a hypothesis. We use $err_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{I}(h(x_i) \neq c^*(x_i))$ to denote the training error and $err_D(h) = \mathbf{Pr}_{x \sim D}(h(x) \neq c^*(x))$ to denote the true error. Recall the theorem from class:

Theorem 1 If $\mathcal{H}$ is finite, in the realizable case

$$m \geq \frac{1}{\epsilon} \left[\ln(|\mathcal{H}|) + \ln\left(\frac{1}{\delta}\right) \right]$$

labeled examples are sufficient so that with probability at least $1 - \delta$, all $h \in \mathcal{H}$ with $err_D(h) \geq \epsilon$ have $err_S(h) > 0$ or equivalently if $err_S(h) = 0$, $err_D(h) \leq \epsilon$.

- (a) **Short Answer:** What does PAC learning stand for? What is the correspondence between ϵ, δ and the full name?

- (b) **Short Answer:** Briefly explain what is the realizable case and what is the agnostic case. Why is Theorem 1 not meaningful in the agnostic case?

- (c) **True or False:** The true error, $err_D(h)$, of any hypothesis h is an upper bound on its training error, $err_S(h)$ on the sample S . Explain your answer.

- ☐ True
☐ False

- (d) **Short Answer:** Briefly explain in 2-3 sentences the importance of sample complexity for machine learning.

5. **True or False:** The VC dimension of a finite concept class H is upper bounded by $\lceil \log_2 |H| \rceil$.

☐ True

☐ False

6. **True or False:** The VC-dimension of the set of quadratic separators

$$H = \left\{ h : \mathbb{R}^d \mapsto \{-1, 1\} \mid h(x) = \text{sign} \left(a_{0,0} + \sum_{i,j}^d a_{i,j} x_i x_j \geq 0 \right), a_{i,j} \in \mathbb{R} \right\}$$

over $x \in X \subset \mathbb{R}^d$ has VC-dimension $\mathcal{O}(d^2)$.

☐ True

☐ False

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